

Bank Churn Analysis using R

Kukoyi Temidayo Ifeoluwa

Overview



<u>Problem Statement:</u> "Why are some customers ghosting our bank?"

In recent months, we've noticed that a growing number of customers are closing their accounts and walking away. Are they unhappy with our services, or are competitors winning their hearts?

This project aims to uncover the secret patterns behind customer churn by digging into their demographics, behaviors, complaints, satisfaction scores, and product usage. Let's find out who's leaving, why they're leaving, and what we can do to keep them!

Introduction

Project Setup

In Preparing the R Environment, I;

- **Imported Libraries:**
  - `readxl` → for reading Excel files
  - `dplyr` → for data cleaning and manipulation
  - `ggplot2` → for creating professional charts
- **Purpose:** These libraries help load, clean, analyze, and visualize the churn dataset.

Loading the Dataset

In Importing the Bank Churn Dataset, I;

- Used `read_excel()` to load: `/kaggle/input/bank-churn-analysis/Bank_Churn_Analysis.xlsx`
- Data contains **customer details** (e.g., age, geography, balance, tenure).
- The **target column** is `Exited`:
- `0` → Stayed
- `1` → Churned
- Converted categorical variables (e.g., Gender, Geography, Card Type) into **factors** for easier analysis and plotting.

Purpose of the preprocessing steps

- Factor conversion ensures R recognizes categories like *Male/Female* or *Active/Inactive* properly.
- This makes **plots more accurate** and avoids treating them as numbers.
- By preparing the dataset first, later analysis becomes **cleaner, faster and interpretable**.

KEY METRICS

Code

```
df <- read_excel("/kaggle/input/bank-churn-analysis/bank churn Analysis.xlsx")

summary(df)
```



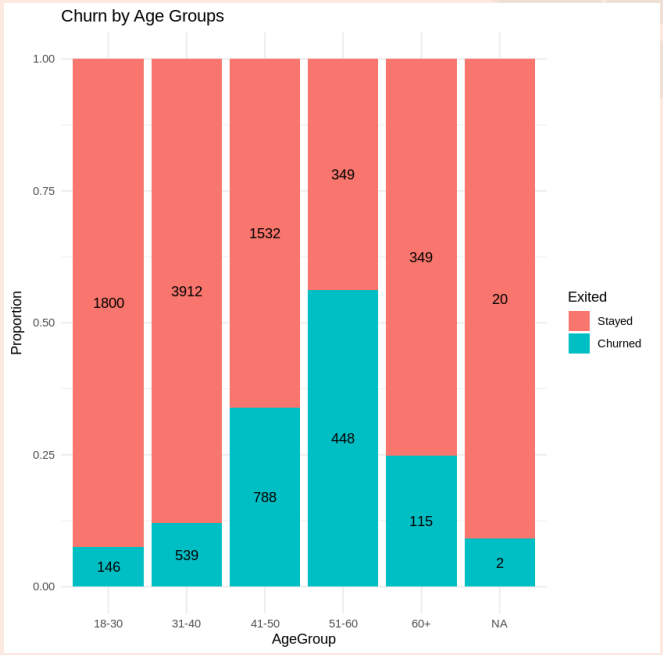
Total Customers	10000
Returned Customers	7962
Churned Customers	2038
Avg Satisfaction Score	3.014
Avg Age	38.92
Avg Tenure	5.013

Churn by Age Groups

Code

```
df %>%
  mutate(AgeGroup = cut(Age, breaks=c(18,30,40,50,60,100),
                        labels=c("18-30","31-40","41-50","51-60","60+"))) %>%
  ggplot(aes(x=AgeGroup, fill=Exited)) +
  geom_bar(position="fill") +
  geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
  labs(title="Churn by Age Groups", y="Proportion") +
  theme_minimal()
```

Output:



Insights

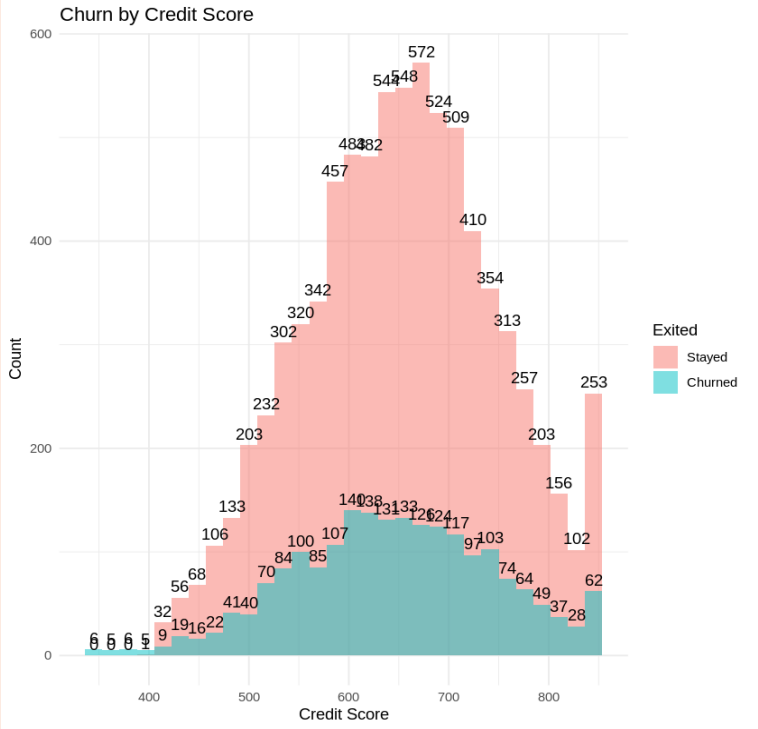
- **What it shows:** Customers in the 51–60 age range churn the most. Younger (18–30) and (31-40) churn less.
- **What it means:** Mid-life customers may be switching banks during career or life changes.
- **Why it matters:** Targeted offers for this age group could help hold them during transition periods.

Churn by Credit score

Code

```
ggplot(df, aes(x=CreditScore, fill=Exited)) +
  geom_histogram(position="identity", alpha=0.5, bins=30) +
  geom_text(stat='bin', aes(label=..count..), bins=30, vjust=-0.5) +
  labs(title="Churn by Credit Score", x="Credit Score", y="Count") +
  theme_minimal()
```

Output:



Insights

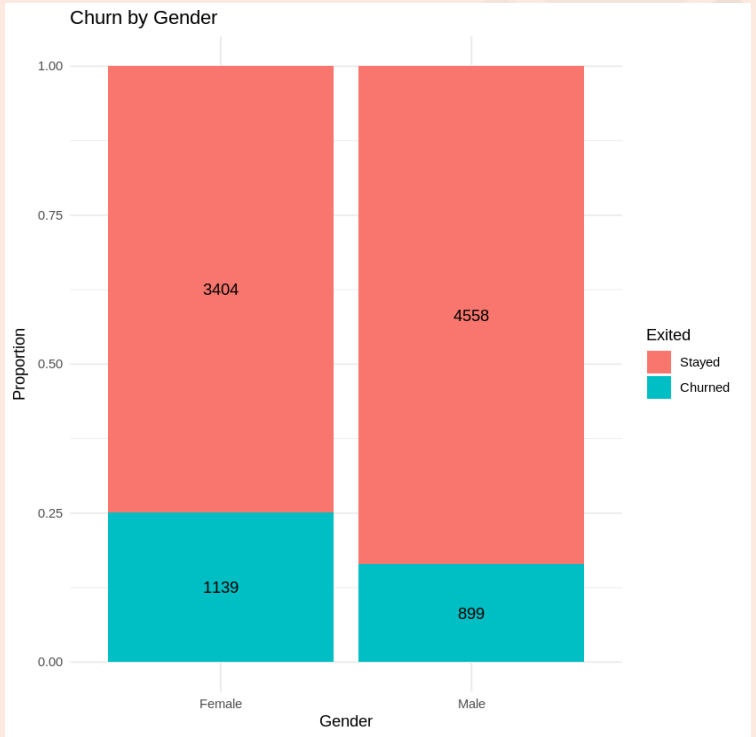
- **What it shows:** Customers with low credit scores (400–600) churn more.
- **What it means:** Credit risk and churn risk are linked. Low-score customers may face more friction (fees, declined applications).
- **Why it matters:** Supporting low-score customers with credit-building products could reduce churn.

Churn by Gender

Code

```
ggplot(df, aes(x=Gender, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Gender", y="Proportion") +
 theme_minimal()
```

Output:



Insights

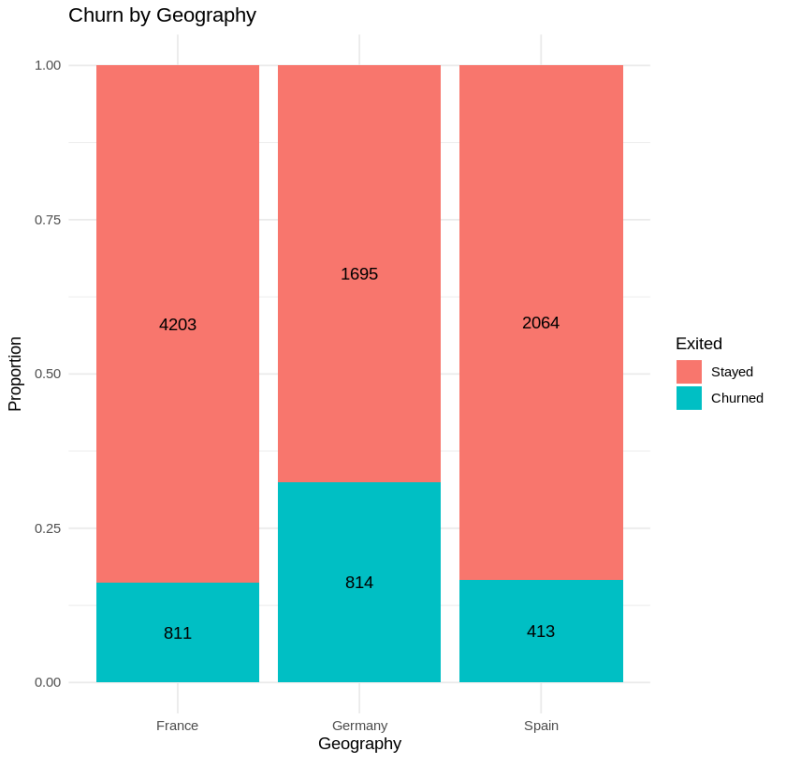
- **What it shows:** Women churn **slightly more** than men. The difference is not huge.
- **What it means:** Gender is a major driver of churn, as the skew exists.
- **Why it matters:** It's worth exploring if specific female customer needs or experiences are not being met.

Churn by Geography

Code

```
ggplot(df, aes(x=Geography, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Geography", y="Proportion") +
 theme_minimal()
```

Output:



Insights

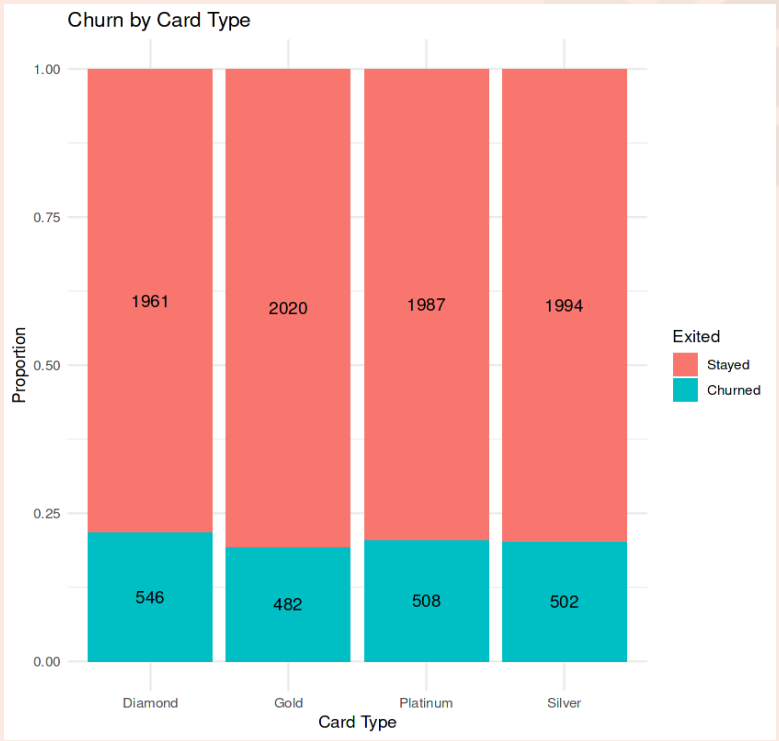
- **What it shows:** Germany has the highest churn. France and Spain have much lower churn rates.
- **What it means:** There are regional differences, possibly linked to service issues, cultural factors, or competition in Germany.
- **Why it matters:** Germany is the “hotspot” for churn and deserves focused retention initiatives.

Churn by Card Type

Code

```
ggplot(df, aes(x=`Card Type`, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Card Type", y="Proportion") +
 theme_minimal()
```

Output:



Insights

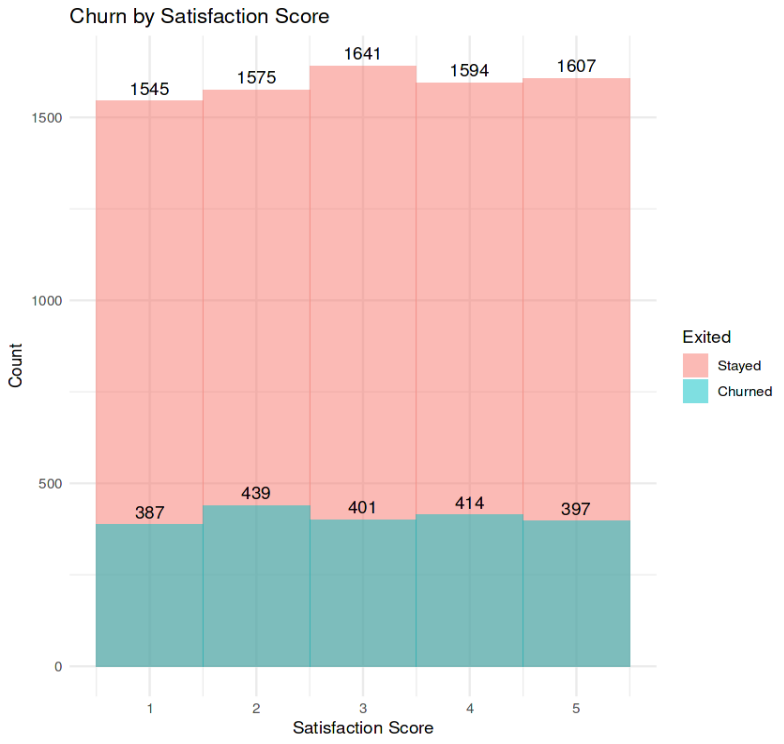
- **What it shows:** Diamond card holders churn more than Gold, Platinum and Silver card holders.
- **What it means:** For Diamond card holders, despite having the top-tier card, they are less satisfied or less engaged than lower-tier members.
- **Why it matters:** Diamond customers may need extra perks or targeted upgrades to reduce their higher churn risk.

Churn by Satisfaction scores

Code

```
ggplot(df, aes(x=`Satisfaction Score`, fill=Exited)) +
 geom_histogram(position="identity", alpha=0.5, bins=5) +
 geom_text(stat='bin', aes(label=..count..), bins=5, vjust=-0.5) +
 labs(title="Churn by Satisfaction Score", y="Count") +
 theme_minimal()
```

Output:



Insights

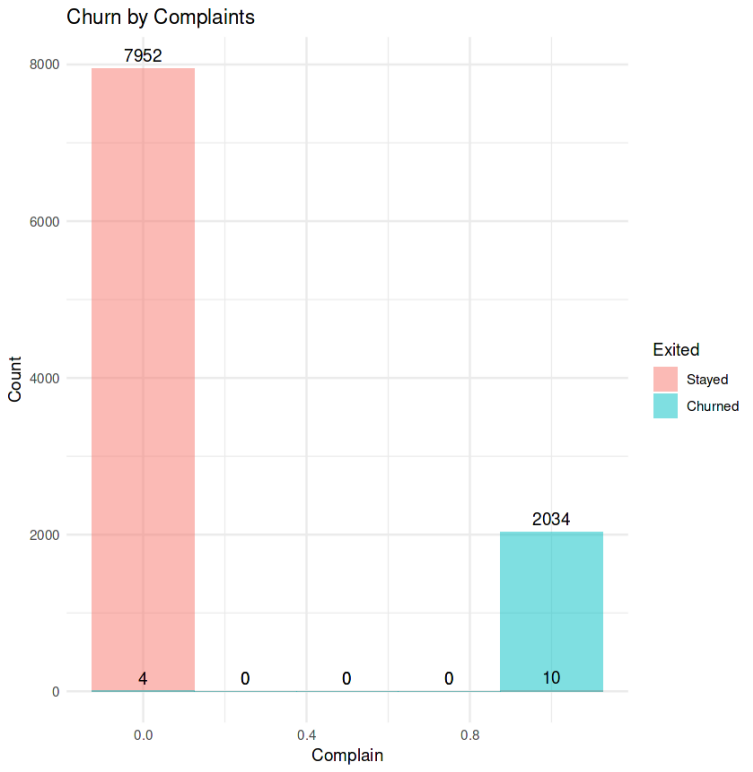
- **What it shows:** Churn rate is fairly flat ~25% across all satisfaction scores, with a small bump at score 2.
- **What it means:** Satisfaction alone is a weak discriminator in this dataset but score 2 shows more churn.
- **Why it matters:** Since churn is almost the same across all satisfaction scores, we cannot rely on satisfaction alone to predict churn and must combine it with other stronger variables.

Churn by Complaints

Code

```
ggplot(df, aes(x=Complain, fill=Exited)) +
 geom_histogram(position="identity", alpha=0.5, bins=5) +
 geom_text(stat='bin', aes(label=..count..), bins=5, vjust=-0.5) +
 labs(title="Churn by Complaints", y="Count") +
 theme_minimal()
```

Output:



Insights

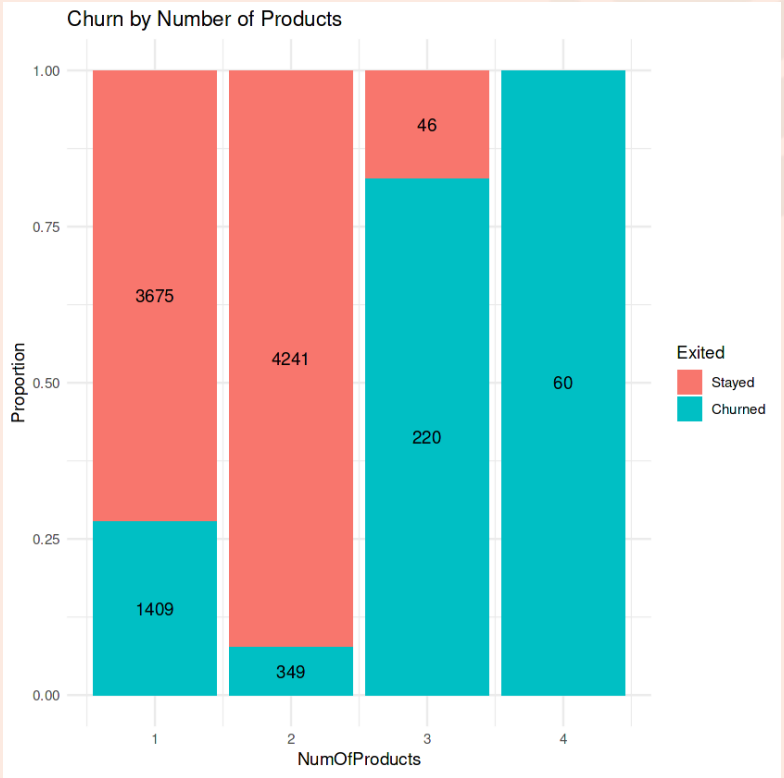
- **What it shows:** Customers who logged complaints churn much more than those who never complained.
- **What it means:** Complaints are a leading indicator of churn risk. Customers who complain and don't get resolution quickly tend to leave.
- **Why it matters:** Complaint management is not just service, it's retention.

Churn by Number of Products

Code

```
ggplot(df, aes(x=NumOfProducts, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Number of Products", y="Proportion") +
 theme_minimal()
```

Output:



Insights

- **What it shows:** Customers with 3–4 products churn the most, while churn is lower for those with 1–2 products.
- **What it means:** Holding too many products may overwhelm customers or signal that they are not finding value in them.
- **Why it matters:** Instead of blindly cross-selling, the bank should focus on right-selling (matching products to needs) and ensuring customers see clear value in each product.

Churn by Credit Card Ownership

Code

```
ggplot(df, aes(x=HasCrCard, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Credit Card Ownership", y="Proportion") +
 theme_minimal()
```

Output:



Insights

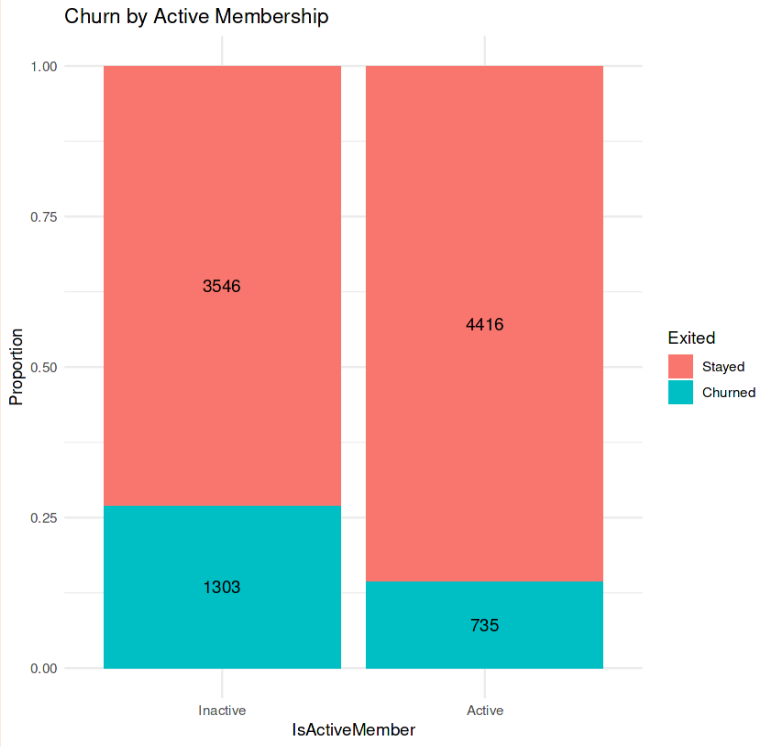
- **What it shows:** Customers with or without a credit card churn at nearly the same rate (about 25%). Technically, customers without a card do churn slightly more, but the gap is almost negligible.
- **What it means:** Having a card may help keep customers engaged, but the effect is small compared to other factors (like complaints).
- **Why it matters:** Credit cards still matter, but they should be seen as one of several retention tools, not the strongest one.

Churn by Active Membership

Code

```
ggplot(df, aes(x=IsActiveMember, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Active Membership", y="Proportion") +
 theme_minimal()
```

Output:



Insights

- **What it shows:** Inactive members churn far more than active members. The gap is relatively significant.
- **What it means:** Inactivity is one of the strongest early warning signs of churn.
- **Why it matters:** Simple activity triggers.(e.g., card swipes, logins) could be used to spot and rescue at-risk customers

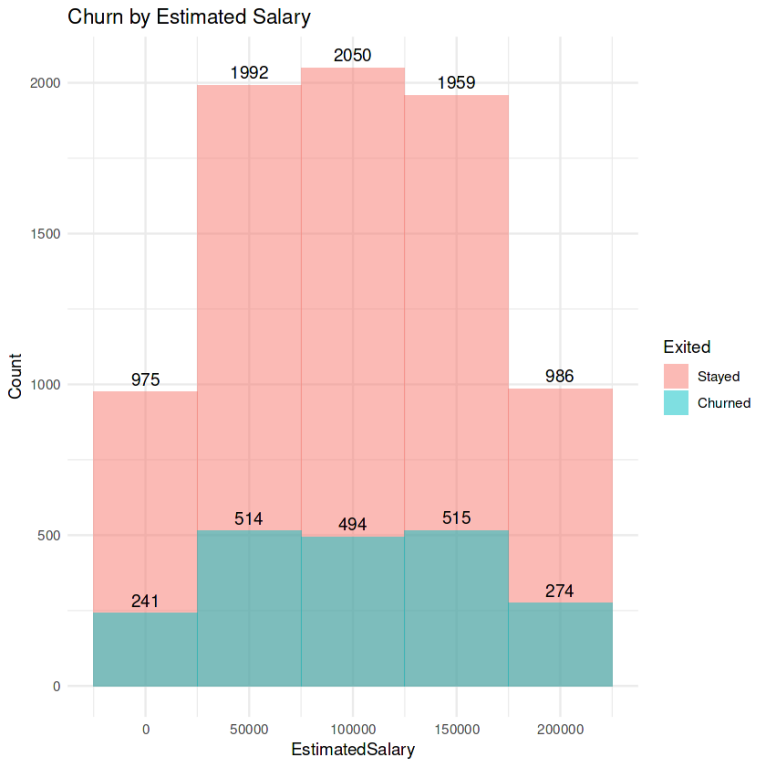
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Churn by Estimated Salary

Code

```
ggplot(df, aes(x=EstimatedSalary, fill=Exited)) +
  geom_histogram(position="identity", alpha=0.5, bins=5) +
  geom_text(stat='bin', aes(label=..count..), bins=5, vjust=-0.5) +
  labs(title="Churn by Estimated Salary", y="Count") +
  theme_minimal()
```

Output:



Insights

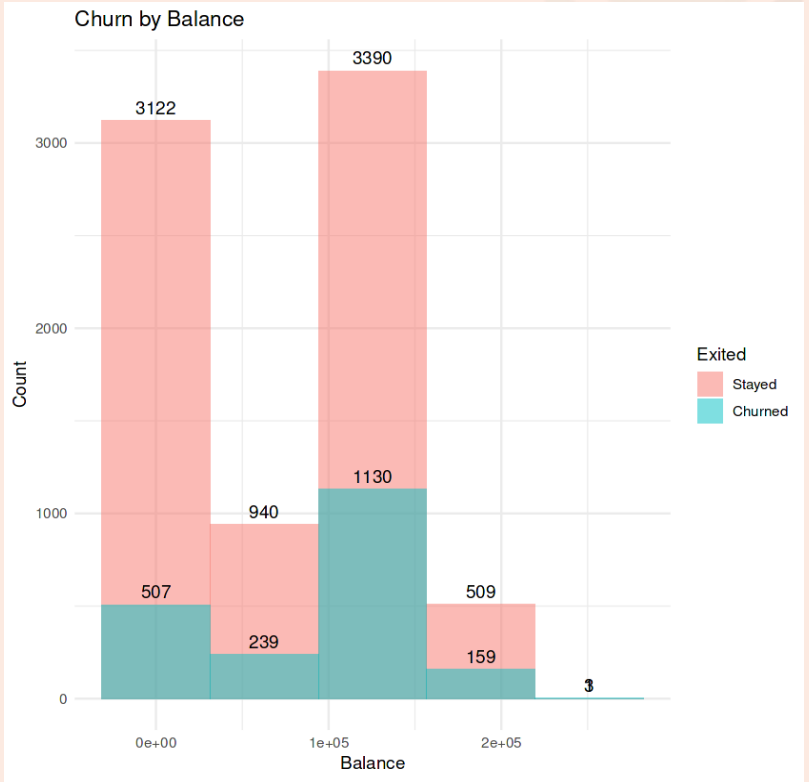
- **What it shows:** Churn rate stays around 20% across all salary levels, with very minor differences (200k+ is slightly higher, but not by much).
- **What it means**: Salary has no meaningful effect on whether customers stay or leave. Churn is more strongly influenced by other factors.
- **Why it matters:** The bank should not assume high earners are safer, loyalty depends more on engagement and experience than income level.

Churn by Balance

Code

```
ggplot(df, aes(x=Balance, fill=Exited)) +
 geom_histogram(position="identity", alpha=0.5, bins=5) +
 geom_text(stat='bin', aes(label=..count..), bins=5, vjust=-0.5) +
 labs(title="Churn by Balance", y="Count") +
 theme_minimal()
```

Output:



Insights

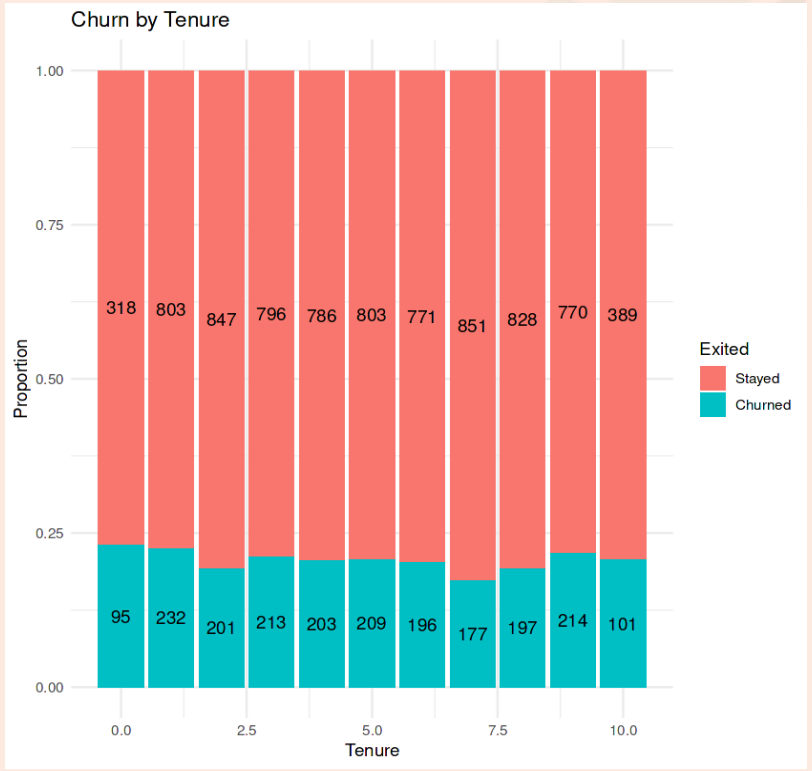
- **What it shows:** Customers with higher-than-average balance churn more often than those with lower balances.
- **What it means:** Having more money in the account does not guarantee loyalty, these customers may be more attractive to competitors or more sensitive to service issues.
- **Why it matters:** The bank must not neglect high-balance customers; proactive engagement and exclusive benefits are needed to stop them from leaving.

Churn by Tenure

Code

```
ggplot(df, aes(x=Tenure, fill=Exited)) +
 geom_bar(position="fill") +
 geom_text(stat='count', aes(label=..count..), position=position_fill(vjust=0.5)) +
 labs(title="Churn by Tenure", y="Proportion") +
 theme_minimal()
```

Output:



Insights

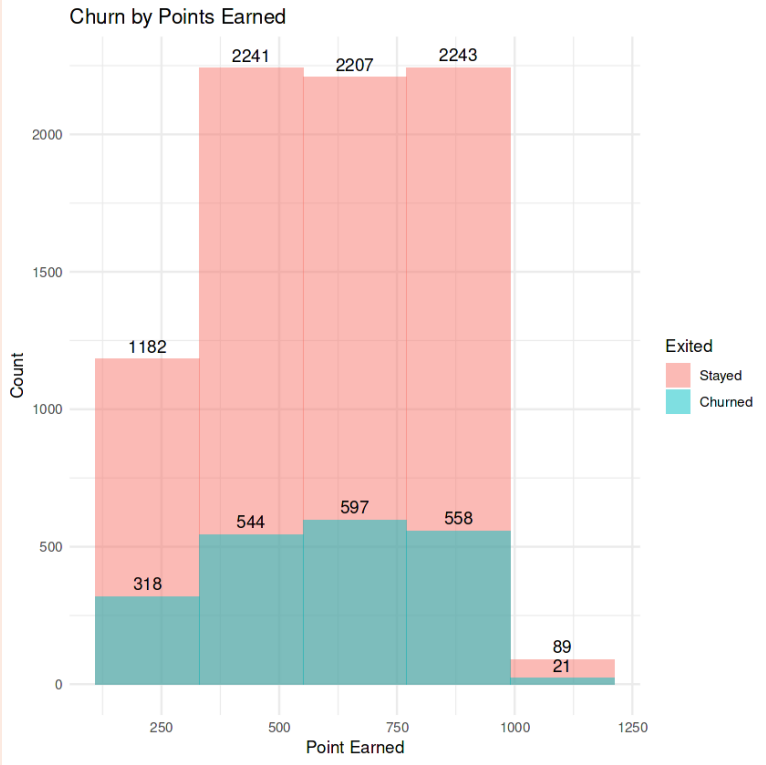
- **What it shows:** Customers in the **3–7 year tenure range** churn a bit more than very new customers (<2 years) and long-tenured customers (7+ years).
- **What it means:** Mid-journey customers are at risk, they've had time to experience the bank but may not yet feel "locked in."
- **Why it matters:** These customers have already invested years with the bank. Losing them here means losing people just before they might have become long-term loyalists.

Churn by Points Earned

Code

```
ggplot(df, aes(x=`Point Earned`, fill=Exited)) +
 geom_histogram(position="identity", alpha=0.5, bins=5) +
 geom_text(stat='bin', aes(label=..count..), bins=5, vjust=-0.5) +
 labs(title="Churn by Points Earned", y="Count") +
 theme_minimal()
```

Output:



Insights

- **What it shows: C**hurn is highest at low points (318 churned vs 1182 stayed), decreases at mid points (544–597 churned vs 2200+ stayed), and is minimal at high points (21 churned vs 89 stayed).
- **What it means:** Customers with low to moderate engagement (fewer points earned) are more at risk of leaving.
- **Why it matters:** Points earned reflect engagement and perceived value. If customers don’t earn enough, they may not feel invested and thus exit.

Recommendations



- Don’t rely only on satisfaction scores, combine churn signals like inactivity, complaints, and product holding.
- Re-engage mid-tenure customers (3–7 years) with targeted loyalty offers.
- Protect high-balance customers with exclusive benefits and proactive care.
- Avoid overselling, match products to customer needs instead of pushing too many.
- Handle complaints faster and track how many customers are saved.
- Focus on Germany with localized retention strategies.
- Support low credit-score customers (400–600) with credit-builder products.
- Add more value for Diamond card holders through upgrade paths or perks.
- Reduce inactivity with transaction nudges, cashback, or digital wallet offers.
- Encourage progression by making it easier to earn points in the early stages, motivating customers to reach higher loyalty tiers.

Conclusion

Churn is mainly driven by **age, activity, number of products and complaints**.

The riskiest customers are **older age groups, mid-tenure, Germany-based, higher-than– average balance, low-credit-score, Diamond card holders, inactive, complaint-heavy and low points earned**.

Acting early on these groups with tailored retention actions will significantly reduce churn.